

Subject: SOFTWARE TESTING	Subject Code: 316314
Semester: 06	Course: CO6K-A
Laboratory No: L003A	Name of Subject Teacher: Prof. Sangeeta Shirsat
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Experiment No:	16
Title of Experiment:	Design test plan and test cases for the login functionality of a social media application.

I. Practical Significance:

This practical focuses on Test Plan which is a document that describes a scope of testing, test strategy, objectives, effort, schedule and resources required. Its main purpose is to guide the whole strategy, objectives, effort, schedule and resources required. Its main purpose is to guide the whole testing process and used mostly by Project Managers or Tests Engineers. The scope of work is defined at the beginning of the testing process. A project team should clearly understand what features and functions to be tested and which ones are out of scope. This practical is useful for students to test any social media application by preparing its test cases and test plan.

II. Practical Related Questions:

1. Write five negative test cases for login with invalid credentials.

ANS:-

Test Case ID	Objective	Steps (Input)	Expected Output	Actual Output	Status
TC_LG_001	Verify login with invalid email format	Enter invalid email (e.g., user@@mail.com) and valid password → Click Login	Error message: “Invalid email format”	Error message: “Invalid email format”	Pass
TC_LG_002	Verify login with incorrect password	Enter valid email and incorrect password → Click Login	Error message: “Incorrect email or password”	Error message: “Incorrect email or password”	Pass
Test Case ID	Objective	Steps (Input)	Expected Output	Actual Output	Status
TC_LG_003	Verify login with empty email field	Leave Email blank and enter valid password → Click Login	Validation message: “Email is required”	Validation message: “Email is required”	Pass
TC_LG_004	Verify login with empty password field	Enter valid email and leave Password blank → Click Login	Validation message: “Password is required”	Validation message: “Password is required”	Pass
TC_LG_005	Verify login with both fields empty	Leave Email and Password blank → Click Login	Validation message: “Email and Password are required”	Validation message: “Email and Password are required”	Pass

2. Define the entry and exit criteria for executing login-related test cases.

ANS:-

Entry Criteria (Preconditions before starting login testing)

1. The social media application is successfully deployed in the test environment.
2. Test environment (browser, OS, network) is properly set up and accessible.
3. Login page UI is fully implemented and functional.
4. Test data is prepared (valid and invalid usernames, emails, and passwords).
5. Test cases for login functionality are reviewed and approved.
6. Required tools (e.g., defect tracking, test management) are ready.

Exit Criteria (Conditions to be met before concluding login testing)

1. All planned login test cases (positive and negative) have been executed.
2. All critical defects related to login functionality are resolved or documented.
3. No severe/blocker defects remain that prevent user login.
4. Test execution results are reviewed and signed off.
5. Test report summarizing passed/failed test cases, issues, and recommendations is prepared.

3. Generate test summary report for the travel booking application.

ANS:

Test Case ID	Test Case Name	Input	Expected Output	Actual Output	Status
TC_FACT_01	Login with wrong username	1. Enter email = wronguser@test.com 2. Enter password = Valid@123 3. Click Sign-	It should show error message: "We cannot find an	It is showing error message: "We cannot find an	Pass



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Test Case ID	Test Case Name	Input	Expected Output	Actual Output	Status
		In	account with that email address"	account with that email address"	
TC_FACT_02	Login with wrong password	1. Enter email = validuser@test.com 2. Enter password = Wrong@123 3. Click Sign-In	It should show error message: "Incorrect password"	It is showing error message: "Incorrect password"	Pass
TC_FACT_03	Login with invalid number	1. Enter email/mobile/username = 935423124011 2. Enter password = abc@5463	It should show error message: "Please enter valid mobile number or create a new account"	It is showing error message: "Please enter valid mobile number or create a new account"	Pass

TC_FACT_04	Login with empty fields	1. Leave email and password blank 2. Click Sign-In	It should show error message: “Please fill the empty fields”	It is showing error message: “Please fill the empty fields”	Pass
TC_FACT_05	Login with valid credentials	1. Enter email = validuser@test.com 2. Enter password = Valid@123 3. Click SignIn	It should show message: “Logged in successfully”	It is showing message: “Logged in successfully”	Pass

Test Plan:

1	Test Plan Identifier	TP_16
2	Introduction:	The purpose of this document is to create a test plan for Amazon Login Module in Mobile Application. The purpose of testing this program is to check the correct operation of its functionality, ease of use.
3	Test Items	Working with the Amazon Login Page
4	Features to be tested	• Login • Sign up • Forgot password

5	Approach	• On the test object: o functional o nonfunctional • According to the requirements o positive o negative • By degree of preparedness - intuitive testing (ad hoc) • Testing using automation tools like load runner
6	Item Pass/Fail Criteria	All test cases with high priority are closed with the result - pass. The test coverage is checked and sufficient, where the criterion of sufficiency is not less than 99% of the coverage of requirements by tests. The test report was compiled and approved by the team lead and customer.
7	Suspension Criteria: Resumption Criteria:	The appearance and entering the bugtracking system of blocking bugs. Closing the blocking bug in the bug tracking system
8	Test Deliverables	Test plan, test case specification, test case ,test summary report
9	Test Tasks	• Writing a test plan • Writing test cases • Development of criteria for the success of testing. • Conducting the testing and evaluation of the results • Creating test reports

10	Environmental needs	Amazon Application Mobile phone Android OS		
11	Responsibilities		Functionality and Responsibilities	Responsible

		<table><tr><td>Login</td><td>Test Engineer 1</td></tr><tr><td>Sign up</td><td>Test Engineer 2</td></tr><tr><td>Forgot password</td><td>Test Engineer 3</td></tr></table>	Login	Test Engineer 1	Sign up	Test Engineer 2	Forgot password	Test Engineer 3
Login	Test Engineer 1							
Sign up	Test Engineer 2							
Forgot password	Test Engineer 3							
12	Staffing and Training Needs	To perform the tasks, you need to have the following knowledge and skills: • knowledge of practical application of the Amazon. • knowledge and ability to apply in practice the basic techniques of test design. • Knowledge of various types of testing including functional and nonfunctional. • Knowledge of automation tool and						
13	Schedule	The deadline for completion of all works and delivery of the project is 14/02/2026 by 5.00pm						
14	Risks and Contingencies Possible risks during testing	• Insufficient human resources for testing the application in deadlines. • Changing the requirements for the product						
15	Approvals Team	Lead Test engineer 1 Test engineer 2 Test engineer 3						

III. Conclusion:

In this practical, we learned how to design a comprehensive test plan and detailed test cases for the login functionality of a social media application. The exercise emphasized understanding the scope, objectives, and strategy of testing, ensuring that all valid and invalid scenarios are covered. This practical helps students develop systematic testing skills, improve software quality, and prepares them to effectively plan and execute tests in real-world projects.

Marks Obtained			Dated Signature of Teacher
Process Related (15)	Product Related (10)	Total (25)	